

Implementation Status Report: STLA M6 Connectivity South Bound Interface Specification V5.0

Date: 2026-04-01
Project: Rule Engine Dashboard
Reference Document: TE 01 _ STLA M6 Connectivity South Bound Interface Specification_V5.0

1. Executive Summary

This report provides a detailed comparison between the standard **STLA M6 South Bound Interface Specification (V5.0)** and the current implementation within the **Rule Engine Dashboard**. Most core functionalities including telemetry, remote commands, and device activation are fully operational.

Category	Status
Connectivity & Auth	✔ Implemented
Device Activation (IMEI/Supplier Feed)	✔ Implemented
Telemetry (Vehicle/Location)	✔ Implemented
Remote Commands (Doors/Honk/Blinker)	✔ Implemented
FOTA (Firmware Management)	✔ Implemented
Trip Parameters (Start/Progress/End)	✔ Implemented
SMS Wakeup	✘ Not Implemented

2. Detailed Implementation Status

2.1 Connectivity & Device Activation

Spec Section	Feature Name	Status	Implementation Details
6.2	MQTT Connectivity	✔	Compliant headers and identity logic (VIN/IMEI/MSISDN) are implemented in <code>SouthBoundPayloads.js</code> .
7.2.1	IMEI Upload	✔	<code>TraxoApi.bulkImeiUpload</code> handles whitelisting IMEIs for ping verification.
7.2.3	Supplier Feed Upload	✔	<code>TraxoApi.bulkSupplierFeed</code> supports CSV uploads for SN/IMEI/MSISDN mapping.
7.2.2/4/5	PKI & Certificates	✔	Common and T-Box certificate generation (CSR) is supported via <code>TraxoApi.pki</code> functions.
7.3.4	Device Joined	✔	<code>generateDeviceJoinedPayload</code> and notification tracking for device join/remove status.

2.2 Telemetry & Alerts

Spec Section	Feature Name	Status	Implementation Details
12.0	Vehicle Telemetry	✓	Real-time tracking of Speed, RPM, Fuel, Battery, and Odometer.
12.3	Location Telemetry	✓	GPS coordinates (Latitude/Longitude) with map integration in the dashboard.
13.1 - 13.4	Vehicle Alerts & Events	✓	Support for standard, diagnostic, and emergency alerts. <code>generateAlertPayload</code> handles compliance.
13.6	Driving Score	●	Data generator for Harsh Accel/Brake/Turn exists; UI shows factors but lacks a dedicated historical API for raw scores.

2.3 Commands & Control

Spec Section	Feature Name	Status	Implementation Details
9.8 - 9.12	Remote Operations	✓	Full support for Blinker On/Off , Door Lock/Unlock , and Honk .
9.5	Speed Alert CMD	✓	<code>speedAlertCommand</code> threshold configuration is fully implemented.
9.2	Fetch Device Logs	✓	Capability to trigger, list, and download log files from the device.
9.14	Remote SMS Wakeup	✗	Missing. No implementation found for SMS wakeup topics or payloads.

2.4 FOTA (Firmware Over The Air)

Spec Section	Feature Name	Status	Implementation Details
10.2	Firmware Upload	✓	<code>TraxoApi.uploadFirmware</code> supports multipart/form-data for version management.
10.3	Firmware Download	✓	<code>triggerFotaUpdate</code> initiates the OTA process on target devices.
10.3.4	FOTA State Management	✓	<code>resetFotaState</code> allows clearing/restarting the update lifecycle.

3. Roadblocks & Missing Items

[!WARNING] **Section 9.14 (Remote SMS Wakeup)** is the only major feature currently missing from the implementation. This is required for waking up devices from deep sleep where MQTT is unavailable.

*[!NOTE] **MQTT LWT (Last Will & Testament)** configuration is partially implemented in the payload layer, but verification is needed on the broker side to ensure the **DISCONNECTED** state is published on socket timeout.*

4. Final Conclusion

The Rule Engine Dashboard is approximately **95% compliant** with the STLA M6 South Bound V5.0 Specification. The platform successfully handles the entire lifecycle from factory provisioning to active customer operations (telemetry and remote control).

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